Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/13/2008
PI Name:	O'Banion, Kerry M.D., Ph.D.		
Project Title:	HZE Radiation Effects on Neuroinflammation: Role of COX-2		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Rochester	**State:**	NY
Zip Code:	14642-0001	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	06/01/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: End date changed back to 06/01/2008 per S. Krenek/JSC (8/07) End date changed to 6/30/2008 per JSC info update (10/06) originally in task book as Michael O'Banion (changed 2/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olschowka, John A (University of Rochester Medical Center) Williams, Jacqueline Patricia (University of Rochester Medical Center) Hurley, Sean (University of Rochester Medical Center)		
Grant/Contract No.:	NNJ04HD82G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Localized activation of glial cells and upregulation of inflammatory mediators, a process called neuroinflammation, is a common feature of brain injury that may contribute to neuropathological processes in diseases ranging from stroke to Alzheimer's. Our laboratory has investigated acute and chronic neuroinflammatory changes following gamma irradiation of the brain in mice. We find dose dependent changes in many indices of neuroinflammation including morphological activation of astrocytes and microglia, induction of pro-inflammatory cytokines and chemokines, upregulation of prostaglandin production, disruption of blood-brain barrier and edema over a time course of hours to days following irradiation. Similar changes are also observed months after injury in association with white matter damage, modeling the clinical entity known as radiation necrosis. Our efforts have specifically focused on the role of prostaglandins in brain neuroinflammation following radiation injury and have revealed that pharmacological inhibition of cyclooxgygenase-2, an inducible isoform first described in our laboratory, attenuates many CNS responses including induction of cytokines and tissue edema. We propose to establish whether similar neuroinflammatory changes occur in mouse brain following exposure to the unique damage imposed by HZE particles. There is a degree of uncertainty in our understanding of the standard radiobiological parameters (e.g. RBE) with respect to the impact of this type of radiation in the CNS. Therefore, in our first aim we will endeavor to use a sufficient dose range in order to compare outcomes with the physiologic and pathologic endpoints established in our current data using gamma radiation. In our second aim, we will use specific inhibitors of cyclooxygenase-2 to determine if prostaglandins play a key role in the neuroinflammatory response to HZE particles. These studies address important fundamental questions about the interaction of HZE radiation with brain tissue and may lead to simple treatment options that provide protection from neuroinflammation and its potential contribution to neurological problems arising in future space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work will help to establish level of risk and possible mitigation of high-LET radiation effects during space travel.
Task Progress:	We have completed all experiments originally proposed in the grant and have largely completed data analysis for the three experiments conducted in Aim 1. The work with COX inhibitors (Aim 2) is still being analyzed; in particular, we need to finalize our results with tissues obtained at early time points. To date, a total of 12 abstracts describing these findings have been presented at national and international meetings; and 3 manuscripts are in preparation. Based on the studies conducted in this grant, the following conclusions can be made: • The neuroinflammatory reaction to HZE particles is different from low LET radiation • There is a biphasic pattern of inflammation-related gene expression at very early (hours) and later time points (1 month) • Early effects include evidence of axonal injury and neuronal DNA damage without clear evidence of apoptosis • Late effects include chronic glial activation, accumulation of peripheral MHC-II positive cells and T cells within the brain parenchyma, and frank loss of white matter volume • Short-term treatment with COX inhibitors does not attenuate the late cell infiltrate phenotype; additional analyses of these experiments are underway With regard to CPR (Critical Path Roadmap) questions, our current work identifies specific neuroinflammatory and micro-lesion effects that are relevant to question 29g. Identification of specific mediators and pathological processes provides a basis for addressing questions 29a and 29b, though additional work will clearly be required. Finally, the identification of specific endpoints following HZE radiation exposure in brain provides quantifiable targets for assessing potential benefits of countermeasures (question 29d), as was investigated in the second aim of this project. A major caveat of our work is the high dose of HZE radiation that was used for most experiments (6 Gy). This dose was selected on the basis of our preliminary dose-finding study as one showing a robust response. In a later dose-response experiment, it is clear that some of the early cytokine changes can be observed with lower doses (e.g. 1 and 2 Gy); however, it is not know whether the late effects would be the same. Obviously the high doses studied here are quite different from those that will be encountered during space flight, both in terms of magnitude and duration. Thus, additional studies will need to be performed to determine if the neuroinflammatory and structural changes observed in the current work apply to more relevant HZE exposures. Similarly, additional work with proton exposure is needed to characterize the neuroinflammatory responses that might be seen following solar flare events. In conclusion, we have provided clear evidence of neuroinflammatory changes in mouse brain exposed to HZE particle irradiation. Further studies, some of which are already in progress, will help to establish whether such changes pose a risk for space explorers.
Bibliography Type:	Description: (Last Updated: 01/07/2026)
Abstracts for Journals and Proceedings	O'Banion MK, Hurley SD, Williams JP, Olschowka, JA. "HZE radiation effects on neuroinflammation: role of COX-2." Presented at Bioastronautics Investigators' Workshop, Galveston, TX, January 10-12, 2005. Proceedings. Bioastronautics Investigators' Workshop, Galveston, TX, January 10-12, 2005. , Jan-2005
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